Wall time per step: Parse 1M PBMC, 500k (346,259 cells)

Seconds per step. Colour is scaled per step and size: fastest framework 0, slowest 1, on square−root seconds.

	bixverse mtx	bixverse h5ad	bixverse GPU*	bixverse MLX*	Scanpy mtx	Scanpy h5ad	Scanpy dask	scrapper mtx	scrapper h5ad	scrapper HDF5	Seurat std mtx	Seurat std h5ad	Seurat BPCells	CyteArc mtx	CyteArc h5ad
load	46	25	21	21	OOM	12	23	OOM	19	372	OOM	OOM	71	223	78
qc	1.8	1.8	1.2	1.2	–	3.8	4.0	–	2.1	5.6	–	–	4.6	0.1	0.1
normalise	n/a	n/a	n/a	n/a	–	3.5	0.0	–	0.4	0.0	–	–	2.3	14	14
hvg	1.1	1.1	0.8	0.8	–	3.2	5.0	–	0.9	13	–	–	14	18	18
pca	2.2	2.0	1.1	0.8	–	5.7	9.8	–	14	51	–	–	352	18	20
harmony	2.1	2.3	1.3	0.8	–	2.4	2.3	–	87	125	–	–	89	153	144
knn_snn	12	12	5.8	5.4	–	31	27	–	27	29	–	–	31	70	76
cluster	4.0	4.1	3.3	3.3	–	14	13	–	85	83	–	–	104	15	15
cluster_fast	12	12	1.9	1.6	n/a	n/a	n/a	–	118	119	n/a	n/a	n/a	n/a	n/a
umap	22	22	2.6	2.5	–	155	153	–	128	127	–	–	198	293	293
dge	2.8	2.9	2.4	2.3	–	602	–	–	39	1,401	–	–	555	23	23
tsne_default	151	154	136	135	–	796	806	n/a	n/a	n/a	–	–	588	n/a	n/a
tsne_fft	39	38	9.4	14	–	109	109	n/a	n/a	n/a	–	–	106	n/a	n/a
tsne_quick	24	25	15	14	n/a	n/a	n/a	–	84	87	n/a	n/a	n/a	n/a	n/a
total (no tSNE)	94	73	39	38	failed	832	237	failed	403	2,205	failed	failed	1,421	828	683

Black: n/a (step not in that framework), OOM or TIMEOUT where the run died, − not reached.
Each step is scaled over the runs that finished it: at 1M the bixverse CPU tSNE is red because the others timed out.
Scanpy dask has no DGE step. bixverse GPU and MLX run natively without a memory cap.
bixverse CPU tSNE builds its kNN with IVF from 500k (kmknn at 100k). (qd) steps are not in the total.